

Minimum Envelopes and Complete Normal-Surface Sectors

Linear ray bounds, exact enumeration, and independent coverage certificates

A research continuation for the ProveIt project

9 October 2026

Abstract

The current ProveIt arrangement backend forms every pairwise equality between triangle potentials and then rejects directions that fail a standard-coordinate extremality test. We identify the smaller geometric object that is actually needed: the common subdivision on which the per-vertex minimum potentials are linear. An anchored cell of this subdivision is linearly isomorphic to a coordinate face of the standard matching cone. Consequently its rays are already standard extreme rays; a subsequent rank filter is unnecessary.

For a compatible sector with k allowed quadrilateral types and matching nullity two, projective normalization reduces the problem to affine lower envelopes on a closed interval. All non-link standard rays are exactly its endpoints and the genuine changes of these envelopes. If p triangle classes and g nontrivial vertex groups survive contraction, their number satisfies $N \leq p - g + 2 \leq 3k + 4$. Empty and one-ray cones are handled exactly. Once the sector kernel is constructed, the implementation uses $O(1 + k + kp + p \log(p + 1) + N(t + p + k))$ exact operations to emit full coordinates, where t is the number of tetrahedra. This simplifies to $O(tk)$ for $k \geq 1$, with native coordinate validation charged separately; bit complexity remains polynomial. We also prove a linear arithmetic verification rule for envelope coverage using one slope-bracket witness per source line. The supplied checker reconstructs the matching model independently and never calls a ray producer.

The package contains an opt-in integration with the maintained implementation, source code, adversarial and independent tests, frozen and fresh Regina comparisons, paired measurements on actual triangulations, and reproducible certificates. The result advances complete candidate enumeration in supplied low-dimensional sectors. A complete sector-family theorem and global control of topology transformations remain necessary for a general $n^{O(\log n)}$ unknot recognizer.

Reading the claims

The fan correspondence, the linear ray bound, and the coverage theorem are proved below. Runtime measurements are experiments on the declared queries. The global recognition corollary is conditional on explicitly stated discovery and coverage assumptions. No general quasi-polynomial recognition theorem is claimed.

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1 The bottleneck and the contribution

The input to the local problem is a finite triangulation and a specified compatible choice of allowed quadrilateral types. The desired output is the *complete* list of standard-coordinate extreme normal rays in that sector, with pure vertex links omitted. This list supports subsequent choices of compressing discs, components, or other normal surfaces. A first positive-Euler witness is a weaker query and must not be substituted for it when comparing algorithms.

The repository snapshot used here is commit `66098968e88bba797143ac1bf7ad0ac4c5f697df`. At that revision, `fastunknot/normal_sector.py` contracts zero-labelled triangle matching edges, constructs a finite consistency matrix, and offers an arrangement method and a support-enumeration method. The arrangement method restricts every coordinate hyperplane and every pairwise triangle-potential equality to the quadrilateral matching space. It enumerates intersections of these hyperplanes, lifts nonnegative directions, and tests whether their positive standard support has a one-dimensional kernel. These steps are exact, but most potential equalities need not describe a minimum. The construction therefore over-refines the geometry before paying for its correction.[\[8, 9\]](#)

Three recent incoming packages were inspected in addition to the maintained code. The dual-certificate package replaces some positive-Euler searches by anchor linear programs and explicitly asks for certified removal of redundant minimum-potential choices. The support-sensitive package concerns the cost of observing supplied normal vectors. The transport package constructs and checks topological changes and ordered boundary data. The present contribution is a complete low-dimensional enumeration method, answering a tractable part of the minimum-potential question while retaining all standard rays.[\[10, 11, 12\]](#)

1.1 What is new in this continuation

The main mathematical statement is a coordinate-face correspondence, valid in all dimensions. It explains precisely which subdivisions preserve standard extremality. Its two-dimensional specialization gives an exact output list and a linear bound; it does not merely bound the number of directions tried by a heuristic. A second structural argument combines the cycle rank of the contracted triangle graph with quadrilateral matching rank to sharpen the initial $4k + 2$ bound to $3k + 4$.

The implementation follows these proofs. It never constructs pairwise potential hyperplanes, never calls the supported-rank filter, and projects triangle potentials only once. The independent checker rebuilds an uncontracted source model from matching equations. A checked slope bracket proves that each original line is above the claimed envelope throughout the interval. Thus a completeness claim can be checked without regenerating the old arrangement or invoking the new producer.

1.2 Relation to established work

Canonical extension from quadrilateral coordinates, removal of vertex links, and general conversion to the standard solution set are established parts of normal-surface theory. Burton gives a canonical extension algorithm and a general conversion algorithm; his work also explains why standard and quadrilateral vertex lists can differ.[\[1\]](#) Affine lower envelopes and their relation to planar convex hulls are classical computational geometry.[\[5\]](#) The contribution here is a proved specialization of canonical conversion, its graph-sensitive count, and its exact integration and validation against the previous implementation. The literature examination does not establish priority for every isolated lemma.

Global bounds obtained from maximal admissible faces provide related context. The closed connected manifold hypotheses in Burton’s asymptotic bounds must be retained; those bounds are not transferred here to finite knot exteriors with boundary.[2]

Linear-programming feasibility and quadrilateral branching also have extensive precedents. Burton and Ozlen study fast branching and tree traversal for normal-surface decisions.[3, 4] We do not describe those general ideas as new. Nor does the present result settle a general output-polynomial conversion question: the bounded-dimensional case is the essential restriction.

Lackenby’s July 2026 preprint supplies a hierarchy-based incompressibility algorithm and analyses hierarchy constructions. Its introduction leaves speed-up to future work; it is not a proved running-time specification for the announced quasi-polynomial goal.[6] The implementation developed here addresses one local part of a prospective certified hierarchy or surface search.

2 Finite normal matching and compatible sectors

Let $\mathcal{T}$ be a finite triangulation of a compact three-manifold, with t tetrahedra. Face identifications and boundary faces are part of the input. The native validation used by the implementation checks its accepted finite-manifold schema before any normal-coordinate claim is made. A triangulation supplied to this API is not, solely by being supplied, a certificate that it is the exterior of a particular knot diagram.

Each tetrahedron has four triangular and three quadrilateral normal disc types. A standard coordinate vector consists of their nonnegative counts. Vertices and opposite faces are numbered 0, 1, 2, 3. Quadrilateral types 0, 1, 2 avoid the opposite edge pairs 01|23, 02|13, and 03|12, respectively. A face-pairing permutation lists the images of vertices 0, 1, 2, 3 in that order. For a face and an arc type, matching takes the form

$$x_a + q_i = x_b + q_j, \tag{1}$$

with a missing or forbidden quadrilateral contribution interpreted as zero. These are finite matching equations. We do not replace them by a possibly different system for spun surfaces in an ideal triangulation.

Definition 2.1 (Compatible allowed sector). A sector Σ is a set of k pairs (τ, j) , containing at most one quadrilateral type for each tetrahedron. Quadrilaterals outside Σ are forced to zero. The allowed coordinates may themselves vanish. Thus k is the number of *allowed* types, not necessarily the positive support of an output ray.

Write $q \in \mathbb{R}^k$ for the allowed quadrilateral coordinates and x for triangle coordinates. The standard sector cone $\mathcal{K}_\Sigma$ consists of the nonnegative solutions of the finite matching equations. It is a coordinate face of the unrestricted nonnegative standard matching cone: forcing forbidden quadrilaterals to zero imposes coordinate-face equations. Hence its extreme rays are exactly the unrestricted standard extreme rays that lie in the sector. This observation justifies comparison with a complete Regina standard-ray list filtered by allowed types.

Two concepts of dimension must be kept distinct. A rational linear matching space can have dimension two while its nonnegative part is empty apart from zero, or is only one-dimensional. The implementation first computes the rational matching nullity and then determines the actual feasible section.

t	Number of tetrahedra in the supplied finite triangulation.
k	Number of allowed quadrilateral types, with $k \leq t$.
A	Triangle-graph cycle consistency matrix on the k coordinates.
d	Rational matching nullity $\dim \ker A$.
a	Number of matching edges with a nonzero quadrilateral label.
p	Number of retained triangle classes after zero-label contraction.
g	Number of retained vertex groups; each retained group has at least two classes.
N	Number of emitted non-link standard extreme rays.
H	A bound on binary sizes of rational numbers in the post-kernel computation.

Table 1: Notation. Vertex-link factors removed by the compiler contribute neither to $p - g$ nor to the non-link ray count.

3 The triangle graph, potentials, and encoded size

Construct a graph whose vertices are tetrahedron corners. Every face matching equation connects its two triangle corners. Orient the edge so that

$$x_b - x_a = \eta_e \cdot q, \quad \eta_e \in \mathbb{Z}^k. \quad (2)$$

Loops and repeated edges are allowed. The connected components of this graph are the global vertices of the triangulation. On a spanning forest, propagate integral vector potentials ℓ_c so that a candidate solution has

$$x_c = \alpha_v + \ell_c(q), \quad c \in G_v. \quad (3)$$

The remaining edges give consistency equations $Aq = 0$. An edge with zero label forces its two triangle coordinates to agree identically and can be contracted. The maintained compiler performs this contraction before building its reduced standard cone.

At a vertex whose corners all contract to one class, that class is a free vertex-link factor and can be omitted from the retained kernel. The other classes form groups G_v of sizes at least two. There are p such classes and g such groups. Formula (3) remains valid on them.

Lemma 3.1 (Label and forest counts). *For a sector with k allowed types,*

$$a \leq 4k, \quad p - g \leq a, \quad \text{rank } A \leq a - (p - g). \quad (4)$$

Consequently, with $d = k - \text{rank } A$,

$$p - g \leq a - k + d \leq 3k + d. \quad (5)$$

Proof. An allowed quadrilateral has four sides, one on each tetrahedron face. Each side can contribute to at most one matching-edge label. Therefore the number of edges with a nonzero label is at most the total $4k$ incidences. Cancellations, boundary faces, and repeated identifications can only lower this count.

Before deletion, each global-vertex triangle graph is connected. Delete the nonzero-labelled edges and contract the connected components of the remaining zero-labelled graph. If there are c_v resulting vertices at a global vertex, a spanning tree of its active quotient graph contains $c_v - 1$

edges. The total forest size is $\sum_v (c_v - 1) = p - g$, including zero contributions from singleton groups. This gives the second inequality.

There are exactly $a - (p - g)$ edges outside a spanning forest, counting loops. After forest potentials are assigned, one cycle consistency equation for each of those edges suffices. The rank of the consistency matrix cannot exceed the number of equations in this presentation. Substituting $\text{rank } A = k - d$ proves (5). $\square$

3.1 A useful inherited coordinate bound

The following argument reestablishes the support-kernel height bound used by the existing implementation. It is included to make binary complexity explicit, rather than interpreting normal-disc multiplicity as input length. The bound itself belongs to the preceding sector work.[9]

Lemma 3.2 (Mixed-column minor bound). *Every primitive nonnegative standard extreme ray in a sector with k allowed quadrilateral types has all retained coordinates at most 4^k . Expanding contracted triangle classes back to tetrahedron corners preserves this bound.*

Proof. Write the retained standard matching matrix in triangle and quadrilateral blocks. The triangle block is the transpose of an oriented graph incidence matrix, possibly with zero rows from loops. Every square minor of that block is 0, 1, or -1 . Each quadrilateral column has total absolute sum at most four, since its entries arise from at most four quadrilateral sides.

Expand a square minor along its $r \leq k$ quadrilateral columns. The remaining triangle minor has absolute value at most one. The sum of the absolute expansion terms is bounded by the product of the column ℓ^1 norms, so the original minor has absolute value at most $4^r \leq 4^k$.

Let an extreme ray have positive support of size s . The matching matrix restricted to this support has rank $s - 1$. Choose $s - 1$ independent rows. Its signed maximal minors give a nonzero integral kernel vector, proportional to the positive ray. Each entry is bounded by 4^k . Dividing their common divisor gives the primitive vector and cannot increase an entry. Restoring contracted triangle coordinates simply repeats entries. $\square$

This argument uses the special incidence block, not just a Euclidean row-norm estimate. Normal surfaces can still contain exponentially many discs, but their individual primitive coordinates here need only $2k + 1$ bits. Intermediate rational bases and projective coordinates can be larger; ordinary determinant bounds for the cycle matrix still give polynomial binary size in (t, k) .

4 Canonical lifts and the exact minimum subdivision

Let

$$\mathcal{C} = \{q \in \mathbb{R}^k : q \geq 0, Aq = 0\}. \quad (6)$$

For each vertex group define

$$m_v(q) = \min_{c \in G_v} \ell_c(q). \quad (7)$$

The canonical lift $\mathcal{E}(q)$ has quadrilaterals q and triangles

$$\mathcal{E}(q)_c = \ell_c(q) - m_v(q), \quad c \in G_v. \quad (8)$$

The same definition can include singleton groups; their canonical coordinate is zero. Let L_v denote the normal vertex-link vector, equal to one on the triangle corners at v and zero elsewhere.

Lemma 4.1 (Canonical decomposition). *Every $z \in \mathcal{K}_\Sigma$ has a unique expression*

$$z = \mathcal{E}(q) + \sum_v \beta_v L_v, \quad \beta_v \geq 0, \quad (9)$$

where q is its quadrilateral projection and β_v is its minimum triangle coordinate at v . Every standard extreme ray with $q \neq 0$ is canonical. The extreme rays with $q = 0$ are the vertex links.

Proof. The triangle differences are prescribed by (3); subtracting the minimum triangle value at a vertex leaves exactly (8). This gives existence and uniqueness. If a vector with nonzero quadrilateral part has a nonzero vertex-link summand, the two summands in (9) cannot be proportional, because their quadrilateral projections differ by being zero or nonzero. It cannot span an extreme ray. For $q = 0$, the connected triangle graph gives precisely the independent nonnegative vertex-link factors. $\square$

Canonicity is standard.[1] The next theorem extracts the exact face structure needed by an enumeration algorithm.

Definition 4.2 (Anchored minimum cell). Choose one class $b_v \in G_v$ at each vertex, including any omitted singleton group conceptually. Define the homogeneous cell

$$F_b = \{q \in \mathcal{C} : \ell_c(q) \geq \ell_{b_v}(q) \text{ for all } v \text{ and } c \in G_v\}. \quad (10)$$

Its projective section is $P_b = F_b \cap \{\sum_i q_i = 1\}$. The nonempty cells and their faces constitute the common minimum subdivision.

Theorem 4.3 (Coordinate-face correspondence). *On F_b , canonical lifting is linear and is a linear isomorphism onto the coordinate face*

$$\{z \in \mathcal{K}_\Sigma : z_{b_v} = 0 \text{ for every } v\}. \quad (11)$$

The non-link standard extreme rays are exactly the lifts of extreme rays of the cells F_b , with duplicates removed. Equivalently, they are exactly the lifts of vertices of the nonempty projective cells P_b .

Proof. On F_b , the selected potential attains the minimum, so the lift is given by the linear expressions $\ell_c - \ell_{b_v}$. They are nonnegative and their anchor coordinates vanish. Conversely, a matching vector in (11) has $\alpha_v = -\ell_{b_v}(q)$ in (3). Its nonnegativity gives the defining inequalities of F_b , and the vector equals $\mathcal{E}(q)$. Quadrilateral projection and this linear lift are inverse maps.

Because all standard coordinates are nonnegative, setting selected ones to zero defines a face of the entire standard cone. Extreme rays of a face are extreme rays of the entire cone. Furthermore a nonzero vector in (11) cannot have zero quadrilateral part: one zero triangle at every vertex removes every pure link direction. Consequently the functional $\sum_i q_i$ is positive away from zero on this face, and its unit section is compact. Vertices of that section correspond to its extreme rays.

For the converse, Lemma 4.1 makes every non-link standard extreme ray canonical. Choose one of its zero triangle entries at each vertex. It belongs to the corresponding anchored face and remains extreme there. The inverse linear map yields an extreme ray of F_b , or equivalently a vertex of P_b . $\square$

4.1 Why the global nonlinearity causes no gap

Canonical lifting is generally not additive on all of $\mathcal{C}$. A proof that assumed global additivity would be invalid. Theorem 4.3 uses additivity only within a specified anchor cell. More explicitly, if $\mathcal{E}(q) = z_1 + z_2$, every zero anchor coordinate of $\mathcal{E}(q)$ must be zero in each nonnegative summand. Both summands therefore lie in the same anchored face. Their projections lie in the same F_b , where lifting is linear. This also explains why a hidden vertex-link summand cannot defeat the extremality argument.

An equality $\ell_c(q) = \ell_{c'}(q)$ between two potentials above the minimum need not define a boundary of any F_b . Subdividing along that equality can introduce a direction that is not extreme in the standard cone. The existing rank filter repairs this over-refinement. The new algorithm avoids it at the source.

4.2 Higher-dimensional interpretation

Restrict the coefficient vectors of ℓ_c to the dual of $\text{span } \mathcal{C}$. For a vertex group, their convex hull has a lower normal fan describing which potentials minimize in a given direction. The common refinement of these fans is the lower normal fan of the Minkowski sum of the coefficient polytopes. Intersecting with $\mathcal{C}$ and the projective section gives precisely the minimum subdivision above. This is a useful geometric description, not a claim that its complexity is small in arbitrary dimension. Boundary faces of $\mathcal{C}$ must also be retained.

5 The nullity-two theorem and a linear ray bound

Assume $d = \dim \ker A \leq 2$. The normalization

$$P = \mathcal{C} \cap \{\mathbf{1}^T q = 1\} \quad (12)$$

meets every nonzero ray of $\mathcal{C}$ exactly once. If $d = 0$, P is empty. If $d = 1$, inspecting the sign of a basis vector decides whether P is empty or a point. If $d = 2$ and the sum functional vanishes identically on $\ker A$, then P is empty as well.

In the remaining $d = 2$ case, choose rational vectors u, w satisfying

$$Au = Aw = 0, \quad \mathbf{1}^T u = 1, \quad \mathbf{1}^T w = 0, \quad w \neq 0. \quad (13)$$

They parametrize the affine matching section as $q(s) = u + sw$. Each inequality $u_i + sw_i \geq 0$ gives a lower bound, upper bound, constant constraint, or contradiction. Since w is nonzero with zero sum, it has both positive and negative coordinates. Thus the feasible parameter set is empty, a point, or a bounded interval $[L, U]$.

For a nondegenerate interval, set

$$f_c(s) = \ell_c(u) + s\ell_c(w), \quad h_v(s) = \min_{c \in G_v} f_c(s). \quad (14)$$

A *genuine interior breakpoint* is a point at which h_v fails to be affine on every neighborhood. A tie between identical lines, a crossing above the minimum, and an endpoint-only tie do not qualify.

Theorem 5.1 (Complete envelope enumeration). *If $L < U$, the complete non-link standard ray list is the primitive canonical lift of $q(s)$ at the parameters*

$$\{L, U\} \cup \bigcup_v \{\text{genuine interior breakpoints of } h_v\}. \quad (15)$$

Repeated parameters yield one ray. If P is a point there is exactly one non-link standard ray, and if P is empty there are none.

Proof. Between consecutive genuine changes of any minimum, each h_v is affine and some fixed anchor realizes it throughout the closed interval. These closed intervals, their endpoints, and possible coincident anchor descriptions form the minimum subdivision of P . Its vertices are precisely the parameters in (15). At a genuine breakpoint, at least one minimum switches between different affine functions, so that point is an endpoint of a minimum cell. No point strictly inside one of the intervening intervals is a vertex of a cell.

There are no missing interior quadrilateral-coordinate events: a nonnegative affine function on $[L, U]$ that vanishes at an interior point must vanish identically. Nontrivial coordinate inequalities therefore only supply the two section endpoints. The theorem follows from Theorem 4.3. A singleton section is a vertex of every nonempty minimum cell, giving the point case; an empty section gives no nonzero ray. $\square$

Theorem 5.2 (Linear count with a cycle-rank refinement). *Let r_v be the number of distinct affine pieces of h_v that occur on open subintervals of a nondegenerate section. Then*

$$N \leq 2 + \sum_v (r_v - 1) \leq p - g + 2, \quad (16)$$

$$N \leq a - k + 4 \leq 3k + 4 \quad (d = 2). \quad (17)$$

The empty and single-ray cases are exact as in Theorem 5.1. In particular the loose bound $N \leq 3k + 4$ applies to every sector accepted by the nullity-at-most-two implementation.

Proof. For a given affine function, its region of being at most every other affine function is an intersection of intervals, hence an interval. It contributes at most one open piece to the lower envelope. Equivalently, concavity of h_v forces the slopes of consecutive distinct pieces to decrease strictly. Thus there are exactly $r_v - 1$ genuine interior breakpoints at vertex v . Taking their union can only reduce their total number. Also $r_v \leq |G_v|$, yielding (16). Apply Lemma 3.1 with $d = 2$ to obtain $p - g + 2 \leq a - k + 4 \leq 3k + 4$. $\square$

The $p - g + 2$ bound is sharper whenever the actual contracted graph is available. The coefficient 3 in the geometric bound is an upper bound; no sharpness claim for three-manifold triangulations is made. The abstract envelope count can be attained: on $[0, 1]$, the p functions

$$f_i(s) = \frac{i^2}{2p} - is, \quad 0 \leq i < p, \quad (18)$$

have successive switches at $(2i + 1)/(2p)$, for $0 \leq i < p - 1$. They give $p + 1$ section vertices. This is an example of a potential cone, not a claimed construction of knot exteriors.

5.1 Degenerate cases are part of the statement

The equation $q_1 + q_2 + q_3 = 0$ has nullity two but no nonzero nonnegative solution. The equation $q_1 + q_2 = 0$ in three variables also has nullity two, but its nonnegative cone is just the q_3 ray. These illustrate why checking only rational nullity is insufficient.

Different potential coefficient vectors can restrict to identical affine functions on the section. They must be coalesced after projection. If slopes agree, only the smallest intercept can matter; if both slope and intercept agree, a deterministic representative is enough. A line attaining the minimum only at a single multiple intersection need not appear as a separate piece: its parameter already occurs as a breakpoint of its neighboring pieces. Interior sphere links and boundary disc links behave identically throughout these proofs, because no Euler-characteristic assertion was used.

6 Exact sweep algorithm and cost

For each vertex group, sort the projected affine functions by decreasing slope and remove larger parallel intercepts. Maintain a stack of retained lines with their start parameters. If the next line meets the last line no later than the last line’s own start, remove the last line and repeat. Each line is pushed once and removed at most once. The retained consecutive intersection parameters increase strictly. Restrict the resulting envelope to $[L, U]$, retaining only pieces with a positive-length intersection, and merge all interior starts into the final parameter list.

Listing 1: Mathematical pseudocode for the proved branch.

```
EnumerateStandardEnvelope(T, allowed_types):
  K := exact finite sector kernel
  require dimension(K.matching_space) <= 2
  P := nonnegative section with sum(q) = 1
  if P is empty: return the empty list
  if P is a point: return its primitive canonical lift
  parameterize P by q(s) = u + s*w, L <= s <= U
  project each retained triangle potential once
  events := {L, U}
  for each vertex group:
    hull := exact affine lower envelope
    events := events union genuine hull starts in (L, U)
  for s in sorted(events):
    q := primitive integer direction of u + s*w
    emit full canonical coordinates using cached projected potentials
```

The hull algorithm is familiar; the reason its event list is a *complete standard-ray list without an extremality filter* is Theorem 5.1. The implementation checks normal-coordinate validity of every emitted lift against the native source geometry.

6.1 Reusing the projected lift

Let $q^{\text{prim}} = \lambda q(s)$ be the primitive integral quadrilateral direction. Since $\sum_i q_i(s) = 1$, the scale is $\lambda = \sum_i q_i^{\text{prim}}$. If the cached projected potential is $f_c(s) = a_c + b_c s$, then the integral triangle value is

$$x_c = \lambda \left(a_c + b_c s - \min_{j \in G_v} (a_j + b_j s) \right). \quad (19)$$

There is no need to repeat a length- k dot product for each class and each output ray. Equation (19) produces integers by the integral finite-matching lift; the code checks that no fractional value survives. Expanding the corner-class map then fills the $7t$ coordinates.

Theorem 6.1 (Post-kernel arithmetic and storage). *Given the exact sector kernel, the nullity-at-most-two branch emits all N full standard vectors using*

$$O(1 + k + kp + p \log(p + 1) + N(t + p + k)) \quad (20)$$

rational/integer arithmetic operations and comparisons, in addition to any separately charged native coordinate-validation work. Its working storage, excluding the supplied kernel and the saved output list, is $O(p + k + t)$ rational or integer entries. The bound is polynomial in encoded input size and has no dependence on expanded normal-disc multiplicity.

Proof. There is constant initial work. The section is obtained by scanning k nonnegativity constraints. Projecting each of p potentials onto u and w takes $O(kp)$ arithmetic operations. Sorting and the stack sweeps cost $O(p \log(p+1))$ overall; the total number of stack operations is linear. Sorting the union of at most $p - g$ breakpoints fits the same bound. Primitive normalization costs $O(k)$ rational/integer operations per output. Cached evaluation, vertex minima, and expansion cost $O(p + k + t)$ per output. Their storage is linear in the listed sizes. Rational operations, including gcd and denominator clearing, have polynomial bit cost in their operand lengths.

The graph labels have bounded integral entries and forest paths have length $O(t)$. Cycle coefficients therefore have polynomial height in t . Rational elimination and projective normalization involve determinants of matrices of order at most k and entries of polynomial height. Their binary lengths are polynomial in (t, k) ; forming affine intersections only adds a bounded number of rational operations. Finally Lemma 3.2 bounds primitive output coordinates. This proves polynomial encoded complexity. $\square$

Corollary 6.2 (A quadratic arithmetic bound). *For $1 \leq k \leq t$ and matching nullity at most two, the post-kernel enumeration cost in Theorem 6.1 is $O(tk)$ exact arithmetic operations, and hence at most $O(t^2)$. Kernel construction and native coordinate validation retain their separate charges. This is an arithmetic bound; the corresponding bit-time statement is polynomial, not necessarily quadratic.*

Proof. Every retained group has at least two classes, so $g \leq p/2$. Therefore $p \leq 2(p - g) \leq 6k + 2d \leq 6k + 4$ by Lemma 3.1. Theorem 5.2, together with the empty and one-ray cases, gives $N \leq 3k + 4$. Thus $p, N = O(k)$ for $k \geq 1$. Substitution into (20) gives $O(k^2 + k \log(k+1) + kt) = O(tk)$, since $k \leq t$. $\square$

If $M(H)$ denotes a chosen bound for exact arithmetic on H -bit operands, multiplying (20) by a polynomial arithmetic factor such as $O(H^2)$ gives a conservative bit bound. The reported $O(p \log p)$ sweep is *not* the cost of constructing a sector kernel, validating the triangulation, or writing its N output vectors. Those costs are stated separately in both the theorem and the measurements.

6.2 What the older arrangement spends

For $d = 2$, the old arrangement takes one hyperplane at a time. The number of distinct projected hyperplanes can be much smaller than the number of raw pairs, but it first considers up to $k + \sum_v \binom{|G_v|}{2}$ pairwise or coordinate equations. It can then lift $O(p^2)$ directions and solve a supported-rank problem for each. There is no claim that this upper bound is attained by every triangulation. In the geometric experiments below, eliminating repeated rank tests is substantial even when the number of distinct hyperplanes grows only linearly.

7 Independent certificates of complete coverage

A producer's assertion that it has reached the end of a search is not an independent proof that its list is complete. The supplied certificate records the matching dimension, projective section, an ordered list of active original corner indices at each global vertex, and the full primitive ray list. For every original corner line it also records a slope-bracket index. The checker validates these data from the triangulation without importing the new producer, its hull routine, the sector-kernel builder, or any ray enumerator.

7.1 One-point dominance of a line over a concave chain

Suppose claimed active pieces are $g_j(s) = a_j + b_j s$, for $1 \leq j \leq r$, with strictly decreasing slopes $b_1 > \dots > b_r$. Their consecutive intersection parameters are

$$z_j = \frac{a_{j+1} - a_j}{b_j - b_{j+1}}, \quad 1 \leq j < r. \quad (21)$$

Require $L < z_1 < \dots < z_{r-1} < U$. These conditions define a continuous concave piecewise-affine chain E , with piece g_j between its neighboring switches. They also imply that E is the minimum of the listed active lines on the entire interval.

Lemma 7.1 (Slope-bracket dominance). *Let $f(s) = a + bs$ be any affine line. A minimum of the convex function $f - E$ on $[L, U]$ occurs at:*

1. L , if $b \geq b_1$;
2. U , if $b \leq b_r$;
3. z_j , if $b_j \geq b \geq b_{j+1}$.

Checking $f \geq E$ at the indicated point proves that inequality everywhere on $[L, U]$. Equal slopes admit either adjacent valid choice.

Proof. On piece j , the slope of $f - E$ is $b - b_j$. Because the b_j decrease, these difference slopes increase. If they are all nonnegative, a minimum is at L ; if all nonpositive, at U . Otherwise the change from nonpositive to nonnegative occurs at a switch with the stated slope bracket. A zero slope gives a constant minimum interval and both endpoints are valid choices. $\square$

Theorem 7.2 (Linear arithmetic envelope coverage replay). *Given m source affine lines, a claimed active chain of r distinct source lines, and one supplied slope-bracket index for each source line, equality between the chain and the full lower envelope on $[L, U]$ can be verified in $O(m + r)$ exact arithmetic operations and comparisons. After the source model and projective section are verified, applying this check to every vertex and comparing the endpoint/breakpoint lifts certifies the complete standard-ray list.*

Proof. Check that every active index names a source line, that no index is repeated, that consecutive slopes decrease strictly, and that the intersections satisfy the strict interval order in (21). These are $O(r)$ checks. For each source line, verify its claimed slope bracket using at most two slope comparisons and apply Lemma 7.1 at the named switch or endpoint. This requires $O(1)$ operations per source line and proves every source line lies above the chain. Conversely, on each chain piece the value is that of an actual source line, so the full source minimum is at most the chain. Equality follows.

Thus the verified chains have exactly the true genuine breakpoints. Theorem 5.1 identifies their union and the endpoints with all standard rays. Independently reconstructing and checking the primitive lifts at exactly these parameters proves both inclusion and coverage. Omitting a genuine switch leaves some source line below the claimed chain and is rejected; inserting a non-genuine switch violates strict slope order, intersection order, or a dominance check. $\square$

7.2 Source reconstruction and the trust boundary

The checker reads native matching rows and constructs an *uncontracted* triangle graph on all $4t$ source corners. It identifies the two triangle coefficients of each row and propagates integral

potentials on a spanning forest. A row whose triangle terms cancel becomes a pure quadrilateral constraint. Residual edges give the finite cycle equations. An older, independent rational eliminator computes their nullity. This model is built without importing the producer's contraction or potential code.

The producer may use a different forest gauge. Two potential assignments with the same differences differ by a common affine function at each vertex; minimum comparisons and slope brackets are unchanged. The certificate therefore identifies original corner indices rather than relying on the producer's private class numbering. It supplies u, w in the common quadrilateral coordinate order, and the checker verifies matching, normalization, independence, and the exact feasible bounds independently.

The $O(m+r)$ coverage bound is for projected-line checking. Reconstructing the graph, certifying matching dimension, projecting its potentials, and checking full output coordinates are additional polynomial work. In the implementation $m = 4t$, since the independent checker deliberately uses all source corners. It makes exactly one dominance test per original corner in a nondegenerate section, including duplicate or inactive lines.

The shared trust boundary is the native finite-triangulation and normal-vector validation, integer transport, and the previous independent rational eliminator. This is not a fully independent implementation of three-manifold topology, and the certificate proves nothing about a diagram-to-exterior correspondence that has not been supplied separately.

8 A geometric family and a non-quadrilateral disc ray

The main growing family consists of a Fibonacci layered solid torus followed by attachment of a tetrahedral ball along one boundary face. The attachment preserves the underlying solid torus while adding a boundary vertex. The generator uses explicit face-pairing tables and integer arithmetic; it does not construct an arbitrary system of linear equations and call it a triangulation. The inherited layered torus fixture supplies a primitive meridian with exponentially growing normal coordinates.[8, 9]

Allow the meridional quadrilateral type in every torus tetrahedron and one chosen type in the attached tetrahedron. This produces matching nullity two even as the number of allowed types grows with the triangulation. In the breakpoint branch the standard list has three rays, while its quadrilateral list has two. The extra ray occurs at the genuine change of the vertex's minimum potential. Independent component checking identifies it as an essential disc in the examples recorded in the package. It is therefore a geometric candidate that would be lost by silently replacing full standard enumeration with the two quadrilateral endpoints.

8.1 An exact all-size classification

Number the $n \geq 1$ original tetrahedra $0, \dots, n-1$. For $0 \leq i < n-1$, pair face 0 of tetrahedron i with face 2 of tetrahedron $i+1$ by $(2, 1, 3, 0)$, and face 1 of tetrahedron i with face 3 of tetrahedron $i+1$ by $(0, 3, 1, 2)$. Pair face 0 of tetrahedron $n-1$ with its own face 1 by $(1, 2, 3, 0)$. The boundary faces are 2, 3 of tetrahedron 0. These are the base Fibonacci solid-torus face maps. Attach the new tetrahedron n to face 2 of tetrahedron 0, identifying its face 2 by the identity permutation. Allow type 2 in each original tetrahedron and type 1 in the cap. Let $F_0 = 0$, $F_1 = 1$, $F_{j+2} = F_{j+1} + F_j$, and put $A_n = F_{n+1}$ and $B_n = F_{n+2}$.

Theorem 8.1 (Capped Fibonacci sectors). *The quadrilateral matching cone of this sector is a two-dimensional quadrant, parametrized by $(x, y) \geq 0$. For integral parameters, put*

$$h = \max\{0, y - A_n x\}. \quad (22)$$

Its canonical normal vector has original-tetrahedron rows

$$(F_{n-i+1}x + h, F_{n-i+1}x + h, h, h; 0, 0, F_{n-i}x), \quad 0 \leq i < n, \quad (23)$$

and cap row

$$(A_n x + h - y, B_n x + h, 0, h; 0, y, 0). \quad (24)$$

There are exactly three primitive non-link standard rays, at

$$(x, y) = (1, 0), \quad (1, A_n), \quad (0, 1). \quad (25)$$

The first two are essential discs. The third is an inessential disc. More generally the canonical integral surface in (23)–(24) consists of x essential meridian discs and h inessential discs; in particular its component count and Euler characteristic both equal $x + h$.

Proof. The type-2 specialization of the base matching equations has triangle coordinates (X_i, X_i, Y_i, Y_i) and quadrilateral coordinate W_i satisfying

$$\begin{aligned} X_i &= X_{i+1} + W_{i+1}, & Y_i &= Y_{i+1}, & Y_i + W_i &= X_{i+1} \quad (0 \leq i < n-1), \\ X_{n-1} &= Y_{n-1} + W_{n-1}. \end{aligned}$$

Starting with $x = W_{n-1}$ and the common offset $b = Y_{n-1}$, these equations give

$$W_i = F_{n-i}x, \quad X_i = F_{n-i+1}x + b, \quad Y_i = b.$$

Conversely these formulas satisfy all base matching equations by the Fibonacci recurrence, so they describe every base solution.

Write the cap triangle coordinates as (c_0, c_1, c_2, c_3) and its allowed quadrilateral coordinate as y . Its three attaching-face equations are

$$c_0 + y = A_n x + b, \quad c_1 = B_n x + b, \quad c_3 = b.$$

The cap triangle at vertex 2 misses the attaching face; write $c_2 = z$. All nonnegative standard matching solutions are therefore exactly

$$x, y, z \geq 0, \quad b \geq \max(0, y - A_n x).$$

The cap introduces one new global vertex, whose canonical minimum is $z = 0$. At the old vertex the minimum triangle entry is $b + \min(0, A_n x - y)$, so canonicity imposes $b = h$ in (22). This proves the coordinate formulas.

The two minimum cells are separated by $y = A_n x$. Their extreme directions are precisely (25), by Theorem 4.3. They are primitive: the last base quadrilateral equals x , which is one on the first two rays, and the cap quadrilateral equals one on the last. The middle direction is interior to the quadrilateral quadrant and hence is not a quadrilateral extreme ray.

For the stated face maps, the type-2 vector with $x = 1$ and $b = 0$ is the primitive essential meridian disc of the preceding dual-certificate report.[10, Proposition 6.4] The support matching cone is one-dimensional and its last quadrilateral coordinate is one, which forces connectedness.

Its Euler characteristic is one and its boundary is nonempty, so it is a disc. Its boundary-edge intersection counts are $F_{n+1}, F_{n+2}, F_{n+3}$; they are not all even, showing a nonzero boundary class over $\mathbb{F}_2$ and hence essentiality on the boundary torus.

For general parameters, uniqueness of normal coordinates identifies the base restriction with x parallel meridian discs and h disjoint small vertex-link discs chosen near the vertex. The attaching face has three distinct boundary-edge classes: this holds in the one-tetrahedron core and persists under the stated layerings. Thus its normal arcs give pairwise disjoint embedded intervals, even though the face closure need not embed. Each present cap triangle or quadrilateral meets this face in one such arc; opposite cap triangles are absent. Each cap disc therefore attaches to one base component along one boundary interval. It preserves that disc and cannot join components. Exactly $x + h$ discs remain.

Essentiality is preserved as well. On the abstract attaching face, the attachment replaces a triangle disc by the complementary abstract boundary disc of the tetrahedral ball, with the same boundary quotient identifications. Each old normal arc is replaced by a disjoint arc with the same endpoints and pairing. Two systems of properly embedded arcs in a disc with the same endpoint pairing are isotopic relative to its boundary. The resulting homeomorphism relative to the abstract boundary respects its quotient identifications and descends to the boundary torus; no embedded closure of the original one-vertex face is assumed. Thus the old and new boundary multicurves correspond. The x meridian boundaries remain essential, and the h link-disc boundaries remain inessential. $\square$

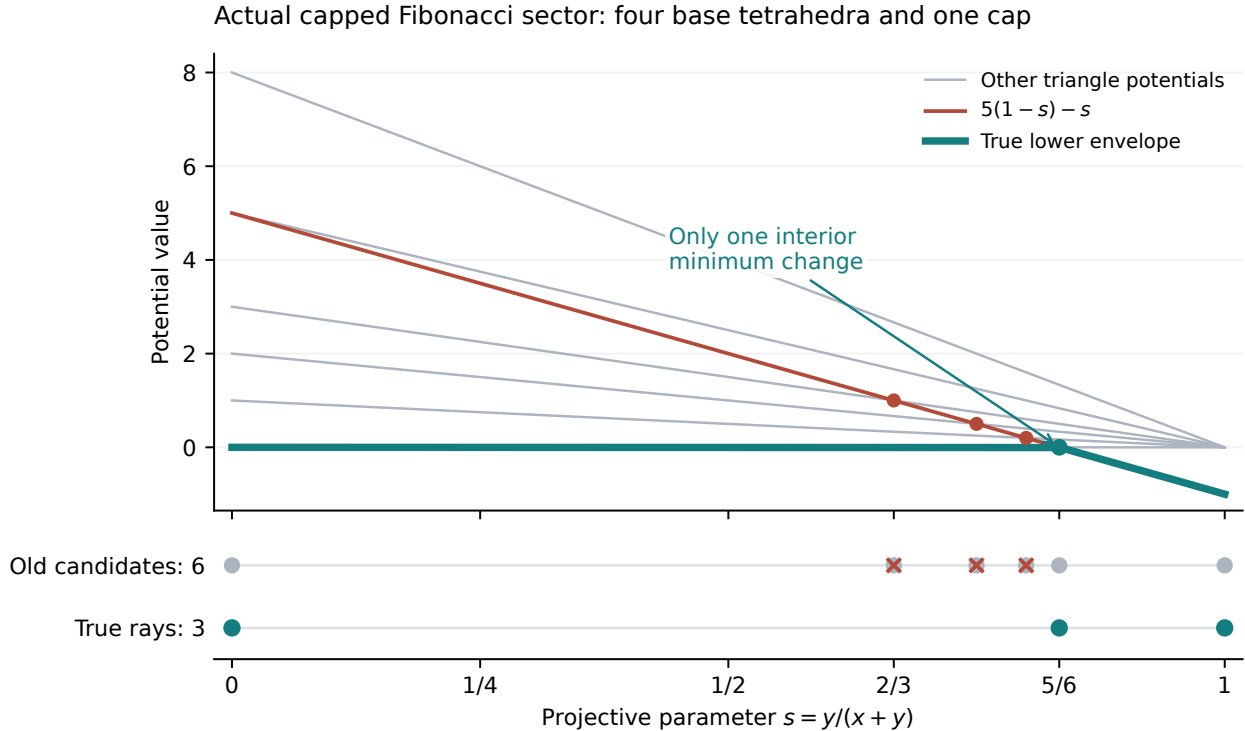


Figure 1: The actual potential functions for the capped family with $n = 4$, using the positive projective normalization $x + y = 1$. Three interior pairwise crossings fail to change the minimum and give no standard ray. The true change at $s = 5/6$, together with the endpoints, gives all three rays. The lower strip marks the complete old candidate list and the exact new output list.

Corollary 8.2 (Exact surplus in the old arrangement). *For this family, the old arrangement has $n + 3$ distinct projected hyperplanes, $n + 2$ nonzero nonnegative candidate directions, and $n - 1$ nonextreme directions that are rejected. The envelope method emits exactly three rays.*

Proof. Modulo a common old-vertex potential, the distinct potential forms are

$$0, F_2x, F_3x, \dots, F_{n+2}x, A_nx - y.$$

The new vertex contributes no differences. Equalities between pure multiples of x all give $x = 0$. The remaining equalities give $y = (A_n - c)x$ for $c \in \{0, F_2, F_3, \dots, F_{n+2}\}$. There are $n + 2$ distinct such slopes, exactly one negative, arising from $c = B_n > A_n$. The quadrilateral coordinate hyperplanes add none: $x = 0$ already occurs and $y = 0$ is the case $c = A_n$. Thus there are $n + 3$ distinct hyperplanes, of which $n + 2$ give nonnegative directions. Theorem 8.1 leaves only three standard rays, so $n - 1$ positive directions fail extremality. Their intermediate equalities do not change the true minimum. $\square$

For $n = 1$, the three rows of the following table give the complete output. The separating semicolon distinguishes triangles from quadrilaterals. The second ray has an independently replayable disc certificate in `examples/extra_non_q_disc.json`.

(x, y)	Original tetrahedron; cap tetrahedron	Topology
$(1, 0)$	$(1, 1, 0, 0; 0, 0, 1); (1, 2, 0, 0; 0, 0, 0)$	Essential disc
$(1, 1)$	$(1, 1, 0, 0; 0, 0, 1); (0, 2, 0, 0; 0, 1, 0)$	Essential disc
$(0, 1)$	$(1, 1, 1, 1; 0, 0, 0); (0, 1, 0, 1; 0, 1, 0)$	Inessential disc

Table 2: The smallest capped-family sector. The interior standard ray is absent from the quadrilateral vertex list.

The formulas also give a direct family-specific oracle using $O(n)$ Fibonacci additions and coordinate operations. The complete output has $\Theta(n^2)$ bits because it contains a linear sequence of Fibonacci coordinates of linearly growing binary lengths. Standard binary addition evaluates the formulas in $O(n^2)$ bit time. The delivered generic kernel builder is not replaced by a special recognizer for this family; the formulas serve as a separate mathematical description and are implemented as an oracle in `experiments/family_oracle.py`.

The example does not claim that the endpoints fail every useful recognition test. In fact the recorded small family also has an essential disc at one endpoint. Its role is to show that the complete output query is stronger than a first-witness query, and to exhibit a large cost of the previous implementation on genuine finite triangulations. The control cap type has no genuine interior change and only two standard rays; it is included to measure the cost when there is less spurious work to remove.

9 Certified reduction to an effective small face

The implementation dispatches on rational nullity at most two. The geometric theorem itself only needs a projective section of dimension at most one. The following elementary certificate can extend applicability after additional preprocessing; automatic production of these certificates is not part of the new code.

Proposition 9.1 (Certified zero-coordinate reduction). *Suppose a rational row vector y satisfies $z = y^T A \geq 0$. Every vector $q \in \mathcal{C}$ has $q_i = 0$ whenever $z_i > 0$. Deleting those allowed coordinates*

leaves exactly the same standard sector cone, with zeros reinserted. If a sequence of such certified reductions leaves matching nullity at most two, the envelope theorem gives a complete ray list for the original sector.

Proof. For $q \in \mathcal{C}$, $0 = y^T Aq = zq = \sum_i z_i q_i$. Each summand is nonnegative, so every summand with $z_i > 0$ has $q_i = 0$. Removing coordinates already forced to zero changes neither the matching solutions nor their standard lifts. The same argument applies successively, and coordinate-face identification preserves the complete standard ray set. $\square$

This is a sufficient certified extension, not a claim that checking individual cycle-row signs finds every forced zero. Linear combinations whose resulting row is nonnegative may reveal zeros that individual rows miss. Exact dual certificates and propagation from the incoming linear-programming work are natural sources for such reductions.[10] Constructing a complete and efficient relative-interior support reduction, and measuring its benefit before envelope enumeration, is a concrete next implementation task.

10 Implementation and integration contract

The new modules use only the Python standard library. Regina is an optional independent oracle for experiments; it is not called by the producer or the certificate checker. Exact integers and `fractions.Fraction` are used throughout. No floating-point decision determines feasibility, equality, breakpoint order, primitive scaling, or completeness.

Module or artifact	Responsibility
<code>sector_envelope.py</code>	Feasible projective section, exact lower envelopes, cached lifts, and the complete ray iterator.
<code>sector_envelope_certificate.py</code>	Compact coverage-certificate production and a capped complete-enumeration API.
<code>sector_envelope_verify.py</code>	Independent uncontracted source reconstruction, dimension check, slope-bracket replay, and full-list checking.
<code>normal_sector.py</code>	A small integration change adds the explicit <code>method='envelope'</code> option for standard enumeration and forwards the method through the supplied-sector discovery API.
<code>tests/test_sector_envelope*.py</code>	Algebra, geometric comparisons, malformed certificates, budget boundaries, and producer-disabled replay.
<code>sector_envelope_research/</code>	Declared corpus audit, actual triangulation generators, paired timing experiments, and supporting data.

Table 3: Implementation modules and their responsibilities. The supplied snapshot also retains their maintained dependencies.

The option is explicit. Existing `auto`, `arrangement`, and `supports` choices retain their earlier selection rules; the quadrilateral phase also retains its earlier path. The patch therefore does not silently introduce a full standard-ray enumeration into the recognizer’s first-witness route. It offers

a faster complete query where its hypotheses are checked. Matching nullity above two is rejected by this method, allowing a caller to choose another exact backend.

Listing 2: Complete enumeration and independent replay.

```
from fastunknot.sector_envelope_certificate import certify_sector_enumeration
from fastunknot.sector_envelope_verify import verify_sector_envelope_certificate

answer = certify_sector_enumeration(triangulation, allowed_types, max_rays=None)
assert answer['status'] == 'COMPLETE'
assert verify_sector_envelope_certificate(triangulation, answer['certificate'])
# answer['coordinates'] is the full non-link standard-ray list in this sector.
```

10.1 Caps and cancellation

In the envelope iterator, `max_bases` counts output candidates whose full lift has begun; there are no redundant arrangement bases in this branch. If a cap is reached before the next candidate, the existing `SearchLimit` exception is raised. A partial iterator is not an absence certificate. The coverage-producing API checks its `max_rays` allowance against the planned complete list and returns `INCONCLUSIVE` without a certificate if it is insufficient.

The same numeric cap can thus permit different progress in the old and new methods, which is expected: their work units differ. Tests use uncapped complete queries for equality claims. Callback cancellation is propagated by identity, including false-valued callable objects and exceptions whose classes coincide with internal limit or geometry exceptions. Hexadecimal integer transport handles large exact fields without altering process-wide decimal conversion limits.

11 Validation and measured performance

11.1 Correctness evidence

The main regression run completed 1,334 tests successfully in 144.558 seconds on CPython 3.12.14. Three additional focused producer tests were subsequently added for multiple vertex groups, multiple genuine changes, and endpoint-only ties; the final focused run completed all 24 producer and certificate tests. Its result is recorded separately in the accompanying test log. This avoids attributing those later test additions to the earlier full-suite run.

The host reports an AMD EPYC 9V74 processor under Linux on x86-64. `results/environment.json` records the runtime and dependency versions. The paired measurements use elapsed wall time; the processor’s model name is not a claim of exclusive access to the host.

The corpus selects supports occurring in recorded standard or quadrilateral rays, every support with at most two allowed types, and 64 fixed-seed full sectors for each triangulation. For triangulations with at most three tetrahedra, it selects every compatible support. The union is deduplicated before testing. Larger triangulations are not claimed to have had every one of their exponentially many sectors tested. The frozen corpus SHA-256 and source pins are retained in `results/audit.json`.

The 15 fresh Regina runs comprise six original corpus fixtures and nine capped-family inputs with base sizes 1, 2, and 4 and each of the three cap types. Regina reports runtime version 7.4; its installed Python distribution is version 7.4.1.^[7] The new producer and checker use no Regina calls. In the separate independent audit, 66 models across all 48 original triangulations were reconstructed both from the uncontracted graph and by dense standard matching elimination. Their nullspaces and all relative vertex potentials agreed; all 60 eligible complete certificates replayed.

Check	Completed instances
Full maintained regression suite	1,334
Final focused producer and certificate tests	24
Frozen finite triangulations represented	48
Declared selected sectors	9,995
Eligible sectors with matching nullity at most two	9,344
Higher-nullity sectors transparently outside the new branch	651
Emitted ray occurrences compared with the complete frozen oracle	4,853
Interior standard-ray occurrences in those comparisons	804
Empty feasible cones correctly returned	6,276
Fresh Regina complete standard enumerations	15
Independent graph-versus-dense model comparisons	66
Complete certificates in the separate source-model audit	60
Family sizes checked against the closed-form ray oracle	9
Integral family vectors with checked component topology	97

Table 4: Validation counts. Ray counts are occurrences over the selected queries, not counts of globally distinct surfaces. All stated comparisons agreed. The six model-only cases had higher nullity.

Matching nullity	Queries	Empty section	Point section	Interval section
0	5,357	5,357	0	0
1	2,837	890	1,947	0
2	1,150	29	140	981

Table 5: Feasible-section stratification. In particular, nullity two does not guarantee a nondegenerate interval.

The affine hull tests use 120 fixed-seed rational line sets and compare against literal minima at every pairwise crossing and points in the intervening intervals, giving over 5,000 exact sample comparisons. They also cover a rational basis change with denominator $2^{257} + 7$ (258 bits). These checks are paired with complete geometric ray-list comparisons, so a correct hull routine alone is not treated as evidence for an incorrect normal-surface reduction.

The separate family oracle evaluates the closed coordinate formulas without calling the matching kernel or envelope routine. Its ray lists agree at base sizes 1, 2, 3, 4, 8, 16, 32, 64, 128. At the first five sizes it also checks the old hyperplane and rejected-direction counts. For base sizes 1, 2, 4, 8, it checks 97 integral vectors with parameters on both sides of the hinge, including the hinge itself. The native compressed component census and its independent certificate replay give exactly $x + h$ components and x essential discs in every case. These finite checks supplement the all-size proof; they do not substitute for it.

11.2 Controlled complete-enumeration measurements

The retained experiment has 13 geometric cohorts, one warm-up per scope, and five paired rounds in alternating AB/BA order. The eight-tetrahedron base with cap type one also has duplicate same-method A/A arms. There are 280 measured calls and 52 warm-up calls across the two scopes, followed by three new-method-only capacity runs. The immutable predecessor is copied from the

pinned repository revision, rather than recreated by editing the optimized implementation. Every measured output hash agrees between arms, and the timed source-file hashes are checked before and after the run.

Table 6 reports medians in milliseconds. Its speedup is the median of the five *paired ratios*, which need not equal the ratio of the two separately reported medians. The last column isolates enumeration with an already constructed kernel. Full raw calls, order, setup costs, work counters, and source hashes are in `results/benchmark.json`.

t	Cap	Old full (ms)	New full (ms)	Full speedup	Reused speedup
2	1	0.626	0.425	1.51×	2.02×
2	2	0.550	0.324	1.75×	2.28×
5	1	8.020	1.021	7.85×	13.88×
5	2	7.516	0.993	7.70×	13.00×
9	1	56.963	2.548	22.89×	41.01×
9	2	75.142	2.603	30.59×	45.10×
17	1	476.595	6.921	68.86×	167.84×
17	2	526.057	9.083	66.86×	171.14×
33	1	6,063.538	29.023	202.71×	781.17×
33	2	6,308.664	30.404	207.49×	728.34×
9	0	17.964	2.119	8.07×	17.00×
17	0	98.115	7.708	12.62×	35.20×
33	0	569.414	28.207	19.75×	75.50×

Table 6: Complete enumeration on capped Fibonacci solid tori. Here $t = n + 1$ includes the cap. “Full” includes kernel construction and full standard-coordinate emission. Types one and two have three rays; type zero is the two-ray control. Every query has matching nullity two.

For $t = 33$ and cap type one, the old and new full medians are 6.064 seconds and 29.023 milliseconds, with paired speedup 202.71×. The reused-kernel medians are 6.593 seconds and 8.204 milliseconds, with paired speedup 781.17×. These are local complete-enumeration measurements on the declared family, not whole-recognizer measurements.

The growth in redundant work is also visible without a clock. At base size $n = 32$, the old backend constructs 35 distinct hyperplanes, tries 34 nonnegative directions, and rejects 31; the new method lifts precisely three directions. Corollary 8.2 proves these counts for every size. The timing difference also includes the cost of constructing pairwise forms and applying dense supported-rank tests, so it should not be explained only by the ratio of candidate counts.

The same-method duplicate ratios (first call divided by duplicate) have medians 0.969 and 0.974 for reused old/new calls, and 0.794 and 1.037 for full old/new calls. This visible host variability argues against treating small differences or last decimal places as reliable. No confidence interval or power-law fit is claimed.

t	Kernel (ms)	Enumeration (ms)	Total (ms)	Kernel share
65	126.833	28.028	154.861	81.9%
129	926.795	144.873	1,071.667	86.5%
257	7,493.208	395.727	7,888.936	95.0%

Table 7: New-method-only capacity runs with cap type one and three emitted rays. Each row is a single call, not a paired comparison or a median.

At $t = 257$, approximately 95% of the new full call is kernel construction. This identifies a concrete implementation target after removing arrangement work. The capacity rows do not imply a measured speedup at these sizes: the predecessor was not run there.

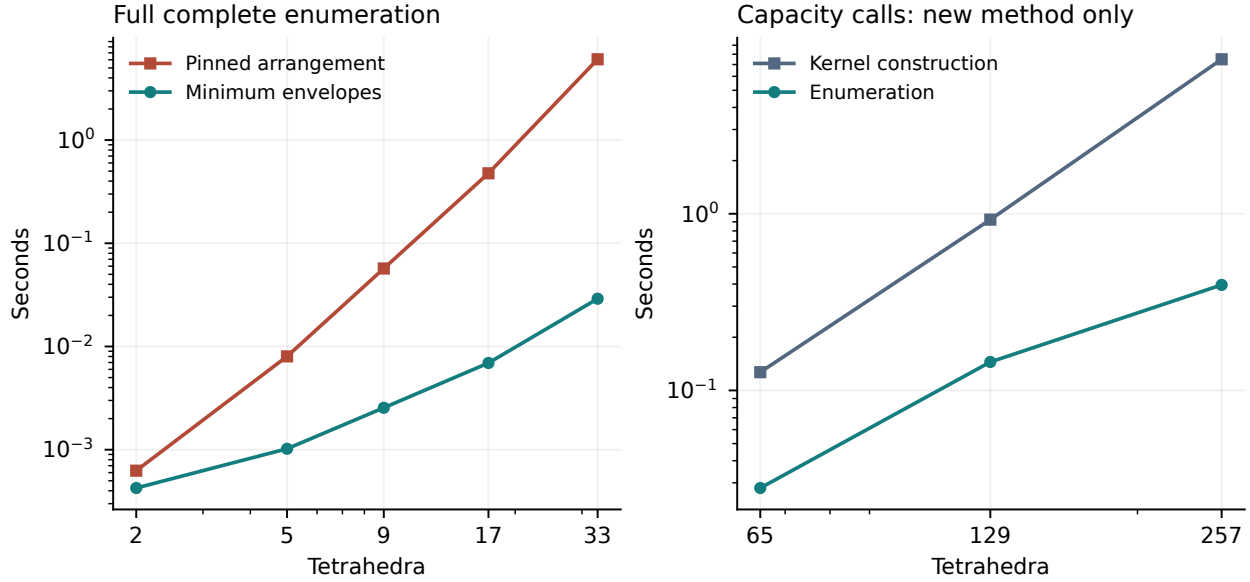


Figure 2: Measured complete-enumeration times. Left: five-round medians of full calls on cap-type-one inputs, including kernel construction. Right: the three new-method-only capacity calls, separated into kernel and enumeration time. Both axes in each panel are logarithmic. Lines connect observations and are not fitted complexity laws.

11.3 Production followed by independent coverage checking

A separate matched workflow includes completeness checking. The old arm constructs the pinned kernel, enumerates with the pinned arrangement, then reconstructs the independent dense matching model and runs its complete reference enumeration to check equality. The new arm produces an envelope coverage certificate and replays it with the independent uncontracted-graph checker. Both workflows require the same complete ray list. The new API constructs its plan for both production and certification; this duplicate constant-factor work is included in the timings.

There is one warm-up and three paired AB/BA rounds for each compared size. Fixture generation and final reporting serialization and hashing are outside the timed regions. The old arm’s ray-list equality check and the new arm’s certificate verification, including their internal hashes, are included. `results/coverage.json` retains every call, source hash, and six independently replayable source/certificate pairs.

t	Old production + replay (ms)	New production + replay (ms)	Paired speedup
2	1.713	1.010	1.68×
9	130.931	8.628	15.18×
17	1,326.585	29.259	51.92×

Table 8: Complete production and independent coverage, cap type one. Reported times are medians; speedups are medians of paired ratios.

t	Production (ms)	Replay (ms)	Total (ms)	Certificate bytes	Tests
33	43.729	53.800	97.529	3,915	132
65	248.320	328.658	576.978	9,985	260
129	1,616.358	2,322.612	3,938.970	29,654	516

Table 9: New-only production and coverage capacity calls. “Tests” counts slope-bracket dominance checks, exactly $4t$ per row. Each list has three rays. JSON sizes include the complete coordinates; serialization is untimed.

The coverage workflow at $t = 17$ has median paired speedup $51.92\times$; its three ratios range from $34.61\times$ to $60.72\times$. This broader scope shows that the benefit persists after checking a complete answer. At $t = 129$, new production and replay take a single combined 3.939 seconds; this is a capacity observation without a comparative claim. The $4t$ dominance count does not imply a linear-time full checker, since matching elimination, projection, and coordinate validation must also be performed.

The audit distinguishes a test method from the thousands of mathematical instances checked inside it. The frozen-corpus selection rule is encoded in the script before results are classified: it is not a list chosen after observing which sectors are fast. Comparisons filter a complete standard-ray oracle by allowed quadrilateral support, which is valid because the sector is a coordinate face. Pure vertex links are removed from both lists.

Fresh oracle comparisons use Regina’s own complete standard enumeration on small declared inputs. They complement, rather than replace, the broader frozen comparisons. Random affine-line tests inspect every pairwise crossing and points in the intervening intervals; they are a check of the hull implementation, whereas the finite-triangulation comparisons test the mathematical connection to normal surfaces.

Timing comparisons use the same exact output query and verify equality of primitive ray sets. Reused-kernel measurements isolate enumeration; rebuild measurements include construction of the matching kernel. Complete coverage measurements additionally run the independent checker. Output multiplicity is never expanded into individual discs for either timing arm.

Ratios are descriptive measurements, not complexity proofs. Repeated rounds and alternating order reduce simple timing bias but do not make the hardware or Python runtime universal. The ordinary recognizer does not automatically call the new complete-sector method, so this report has no measured end-to-end knot-recognition speedup or increase in its recognition coverage.

12 What follows for quasi-polynomial recognition

The local result is useful even for dense supports: k can be proportional to t , and primitive disc coordinates can be exponentially large. The condition concerns the residual matching dimension, not a logarithmic bound on the number of occupied tetrahedra. This avoids the universal sparse-support assumption already contradicted by layered meridians in the previous sector work.[9]

It does not bound how many sectors must be discovered. A fast observer of a supplied normal vector, a fast first positive-Euler query, and a fast complete enumerator inside one specified sector each address different parts of the problem. Only a source-bound construction with a coverage argument can turn those local tools into a complete recognition algorithm.

Theorem 12.1 (Conditional complete-family bound). *Suppose a deterministic procedure on an n -crossing knot diagram constructs at most $S(n)$ supplied finite sectors, possibly after certified transformations, and the following properties hold:*

1. *Source correspondence and every topology transformation are certified and preserve the recognition problem under the stated terminal rules.*
2. *Every generated sector has polynomial encoded size in n and matching nullity at most two, or is reduced to such a sector by checked zero-coordinate certificates.*
3. *Construction and all auxiliary work, including unsuccessful branches, restarts, and terminal checks, cost $S(n) \text{ poly}(n)$ in total.*
4. *For every generated normal vector, a deterministic procedure computes the required component and essential-disc predicates, produces the corresponding source-bound certificates, and replays them in time polynomial in the original diagram size n , under the actual terminal hypotheses of the strategy.*
5. *If the original diagram is an unknot, at least one of the enumerated standard rays contains the required certified compressing-disc witness; the negative conclusion from complete exhaustion is justified by this coverage statement and the source correspondence.*

Then complete enumeration and certified disc search over this family can be performed in $S(n) \text{ poly}(n)$ bit time. In particular, if $S(n) \leq n^{O(\log n)}$, the resulting algorithm has that same asymptotic bound.

Proof. Theorems 5.2 and 6.1 give polynomially many polynomial-size candidates in each sector, with polynomial encoded generation cost. Primitive-coordinate size is bounded by Lemma 3.2, and the independent coverage check is polynomial by Theorem 7.2. Applying the deterministic compressed component procedure in the fourth hypothesis adds polynomial time per candidate, including production and replay of the required source-bound certificates. The hypotheses bound all other work and justify both the positive and negative conclusions. Summing over the at most $S(n)$ sectors yields the claim. $\square$

The final coverage hypothesis is not established here. It is not enough that many experimental discs happen to lie in small-dimensional sectors. The other hypotheses likewise require costs in the original diagram size, not only in the size of an uncontrolled intermediate triangulation. If a hierarchy generates a tree of states, $S(n)$ must include all explored nodes and abandoned suffixes. Polynomial branching through $O(\log n)$ levels gives the desired $n^{O(\log n)}$ count. Replacing that depth by $O(\log^2 n)$ yields a different exponent and cannot be silently conflated with the stated target.

12.1 A precise next milestone

A useful next target is a certified procedure that selects a useful surface from a small-dimensional sector and carries out one complete topology transition, including the ordered boundary attachments needed to continue. The new result supplies complete local alternatives and a checkable exhaustion record for that selection. It still needs a selection policy, an embedding-aware continuation test, and a potential controlling how often the state must be changed or revisited.

13 Further research questions and priorities

1. **Certified effective-dimension reduction.** Implement production of Proposition 9.1 certificates from the incoming dual solver. Measure how often raw nullity above two falls to two on the relative interior support. Compare the complete cost of reduction and enumeration against the current arrangement, rather than reporting the dimension decrease alone.

2. **Minimum subdivisions in dimension three.** For matching nullity three the normalized domain is a polygon. Develop an exact traversal of its actual minimum subdivision, including boundary vertices and coincident faces. The fan theorem removes rank filtering once these vertices are certified, but the arrangement can no longer be replaced by a one-dimensional event merge. Establish explicit output and intermediate complexity bounds for a carefully specified representation.
3. **A geometric bound on complete sector families.** Find a deterministic family of low-dimensional sectors that covers a compressing-disc witness whenever one exists in the relevant knot-exterior class. Identify which triangulation or hierarchy restrictions make such a claim plausible. A bound on the number of sectors is more valuable to the global goal than another improvement to a fixed supplied vector.
4. **Surface choice using complete alternatives.** The extra interior standard rays can have useful topology. Design a component-aware objective that chooses among all rays using essentiality, boundary order, and the complexity of the subsequent cut. A first positive Euler value is insufficient to define this objective. Preserve explicit counterexamples where Euler or aggregate boundary tests discard necessary information.
5. **Incremental kernels under certified local moves.** The envelope stage can become much cheaper than rebuilding the matching kernel. Track how cycle rank, potentials, and active hulls change under the already certified local triangulation operations. Reuse only data whose source correspondence is checked; otherwise a fast stale envelope is an incorrect one.
6. **Persistent coverage certificates.** Determine whether unchanged vertex envelopes can retain their certificates across a local modification. A valid approach must transport corner names, projective parameters, and matching-space information. Prove an amortized bound for all required invalidations, not just successful reuse cases.
7. **Compressed output and streaming topology.** Full $7t$ vectors impose an $\Omega(tN)$ output requirement. Couple the projected-lift representation directly to the compressed component observer while keeping an independently checkable source binding. This could save repeated coordinate scans, but should not hide the cost of any requested full output vectors.
8. **Sharp geometric ray counts.** Is $3k + 4$ asymptotically sharp for realizable finite-manifold sectors with matching nullity two? The abstract lower-envelope construction only shows sharpness of the piece-count expression. Construct a geometric linear lower bound, or exploit additional normal-matching structure to improve the coefficient or the dependence on the number of vertex groups.
9. **Oracle-independent topology validation.** Extend the certificate boundary so that triangulation validity and the diagram-to-exterior construction have independent native replay. The present checker is independent of enumeration, but deliberately shares some native geometry validation with the producer. This is a concrete trust-boundary improvement distinct from faster ray generation.
10. **Whole-search accounting.** Use the new complete local query inside one specified recognition strategy and record every unsuccessful sector, cutoff, repeated state, and topology repair. Seek a decreasing or amortized potential in original-input size. Any quasi-polynomial claim must bound that complete trace, not only the number of successful hierarchy stages.

A Certificate schema and adversarial checks

The schema identifier is `normal-sector-envelope-v1`. It binds the triangulation by its canonical JSON SHA-256 digest and includes an ordered allowed-type list. Exact rational values are reduced numerator–denominator pairs with a positive denominator. Large integer fields can use the repository’s explicit signed hexadecimal representation.

The checker accepts exactly the stated sector coverage claim, not arbitrary metadata attached to it. It verifies the independently recovered matching dimension; emptiness or exact projective bounds; an anchor chain for every global vertex in a nondegenerate section; one checked slope bracket for each original corner; and the complete ordered list of primitive canonical lifts. Matching dimensions above two are outside this schema’s accepted method.

Mutation tests cover a changed source hash, omitted or additional ray, incorrect coordinate, false matching dimension, shortened interval, noncanonical rational, missing or duplicated vertex record, wrong active corner, missing dominance hint, and an incorrect bracket. A specially constructed affine-line adversary places an omitted line below the claimed chain between its endpoints; the slope-bracket test rejects it. Merely checking the global interval endpoints would miss such an undercut.

The producer-disabled test replaces the new enumeration and hull functions, the sector-kernel builder, and reference ray enumerators with functions that raise if called. A previously generated valid certificate still replays. This tests an actual independence boundary, rather than merely checking that the producer agrees with itself.

B Reproduction and integration

The archive includes a self-contained source snapshot for the focused experiments, the exact predecessor of the changed sector dispatcher, a patch against the pinned repository, declared fixtures, raw measurements, and this article’s source. The top-level `README.md` gives the current commands and explains quick and full reproduction. Optional Regina checks require the recorded Regina version; the production code and focused native certificate tests need only the standard library.

The integration patch adds the new modules and research/test files and changes only the supplied-sector method dispatch in maintained code. It does not modify the repository’s historical reports, apply other incoming patches, or replace the recognizer’s default strategy. Applying it to a different revision requires the ordinary review of overlapping changes. The patch is supplied for integration; no remote repository mutation is part of this delivery.

The recorded full-suite result applies to the pinned checkout plus this patch in the recorded runtime. The broad corpus result counts the selected eligible sectors tested, not every sector of every possible triangulation. The timing result concerns the exact queries named in the tables. These three statements should remain separate when the work is incorporated into the maintained synthesis.

C Claim ledger

Claim	Status	Evidence or limitation
Anchored minimum cells are coordinate faces under canonical lifting.	Proved	Theorem 4.3, all dimensions.
Endpoints and genuine lower-envelope changes enumerate every standard ray when matching nullity is at most two.	Proved	Theorem 5.1, including empty and singleton sections.
$N \leq p - g + 2 \leq 3k + 4$ in the nondegenerate nullity-two case.	Proved	Cycle-rank argument and Theorem 5.2.
Primitive coordinate bound 4^k .	Inherited; reproved	Mixed-column minors, Lemma 3.2; preceding sector work.
Post-kernel enumeration has the stated polynomial encoded cost.	Proved	Theorem 6.1; kernel, validation, and full output charged.
One slope-bracket test per original line certifies complete envelope coverage.	Proved	Lemma 7.1, Theorem 7.2.
The implementation produces the declared oracle ray lists.	Checked	Frozen corpus, fresh Regina checks, and focused independent comparisons.
Measured speedups on the declared capped layered-torus queries.	Measured	Paired raw timings; query and preprocessing scope stated.
More diagrams are recognized, or ordinary recognition is faster.	Not established	The new complete-sector method is explicit and is not automatically invoked by the default recognizer.
A complete $n^{O(\log n)}$ algorithm for arbitrary knot diagrams.	Open in this work	Missing complete family and transformation bounds; Theorem 12.1 states the required hypotheses.

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- [11] ProveIt incoming research package, `unknot_support_sensitive_certificates_20261009.zip`. `docs/incoming/`, at the repository snapshot in [8].
- [12] ProveIt incoming research package, `unknot_geometric_transport_20261009.zip`, *Cocycle Transport and Ordered Boundary Interfaces for Unknot Recognition*. `docs/incoming/`, at the repository snapshot in [8].